Alessandro Salo

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PROJECTS

GPU Wavefront Ray Tracer on Vulkan Compute

- Built a Vulkan compute backend from scratch and a fully GPU-parallel LBVH (Karras 2012: Morton codes, radix sort, hierarchy construction, bottom-up AABB propagation) in GLSL compute shaders.
- Aided in converting megakernal ray tracing into five-stage wavefront pipelin.e
- Achieved **352× rendering speed-up** (5067ms to 14.4ms) and a **~9×** faster BVH construction, reaching real-time rendering.

C++20, Vulkan, GLSL/SPIR-V, OpenCL, CMake

Acceleration Structure Comparisons of BVH, BVH-SAH & Grid

- Implemented and benchmarked three CPU ray-tracing acceleration structures (median-split BVH, SAH-optimized BVH, uniform grid with 3D-DDA traversal) across dense, sparse, and uniformly-distributed scenes.
- Instrumented per-ray traversal and intersection statistics to quantify the space- vs. object-partitioning tradeoff.

C++17

SIMD & GPU Cloth Simulation Optimization

- Parallelized a Verlet-integration cloth simulation using SSE intrinsics (SoA data layout, lane-shuffled race-condition-free constraint solving) and OpenCL compute kernels.
- Achieved **5×** (SIMD) and **19.6×** (GPU) speed-ups over the sequential baseline while preserving simulation fidelity.

C++, SSE/SIMD, OpenCL

Rendering Resonance (Rasterizer)

- Built a billboard-based particle system (fire, smoke, fireflies) and normal mapping (tangent/bitangent computation, TBN reconstruction) for a team rasterizer alongside bloom, fog, and toon shading.

JavaScript, WebGL, GLSL

Additional Projects

- **ML & Deep Learning:** CNN classifiers, a from-scratch YOLO detector, and fine-tuned BERTweet for sentiment classification (PyTorch, HuggingFace Transformers).
- **Monte Carlo Simulation Framework:** interactive C++ statistics framework with a custom AST-based math-expression parser.
- **3D Voxel Reconstruction:** multi-camera calibration, background subtraction, and voxel carving pipeline (Python, OpenCV).

EDUCATION

MSc Game & Media Technology - *Utrecht University* 2025 – present

- Coursework in graphics, GPGPU, and optimization; current weighted average 8.43/10.
- Advanced Graphics 9.1, Computer Animation 8.8, Computer Vision 8.1; Optimization & Vectorization (10), and more.

MSc Computational Science & Engineering - *EPFL* 2024 – 2025

- Scientific computing in C++, machine learning, and simulation of physical systems.
- Transferred to Utrecht University.

BSc Physics & Astronomy - *Utrecht University* 2020 – 2024

- Cum laude - weighted average 8.64/10.
- Notable courses: Structure of Matter 9.6, Advanced Statistical Physics 9.5, Numerical Analysis 9.0.

BSc Liberal Arts & Sciences (Honours), Major in Mathematics and Physics - *University College Utrecht* 2020 – 2024

- Magna cum laude - GPA 3.87/4.0 (197.5 ECTS).

RESEARCH & EXPERIENCE

Utrecht University - Physics Research 2023 – 2024

Research Project (Physics BSc Thesis) that led to a Peer-Reviewed Publication

- Ran **direct-coexistence simulations** to determine fluid–crystal phase boundaries of a binary hard-sphere mixture.
- Co-authored the resulting peer-reviewed paper in the **Journal of Chemical Physics** (164(12), 2026; doi:10.1063/5.0321591).
- Built and validated the simulation and analysis code in C; thesis graded 8.9/10.

Utrecht University Aug 2022 – Jun 2024

Science Student Assistant (Teaching Assistant) - Utrecht, Netherlands

- Sole student tutor and TA for all University College Utrecht mathematics and physics courses across two years.
- Graded assignments and gave structured, individual feedback to undergraduate students.

SKILLS

Languages: C/C++, Python, C#, Bash, GLSL, OpenCL C, JavaScript

Graphics & Rendering: ray tracing, acceleration structures, rasterization, physically-based &normal mapping, particle systems

HPC & GPU: Vulkan (compute + graphics), OpenCL, SPIR-V, GLSL compute shaders, parallel algorithms, SIMD (SSE / AVX), OpenGL, WebGL

Scientific Computing: numerical methods, Monte Carlo simulation, statistical mechanics, NumPy, CMake

ML & Computer Vision: PyTorch, HuggingFace Transformers, CNNs, deep learning, OpenCV

Tools: Git, Linux, GPU profilers, GoogleTest, Doxygen, ImGui

CERTIFICATIONS & PUBLICATIONS

AI & Machine Learning: Advanced Neural Networks & Deep Learning (Grade A)	*Oxford - Lady Margaret Hall*	2023
Peer-reviewed publication (co-author), direct-coexistence simulations	*J. Chem. Phys. 164(12)*	2026

INTERESTS

Hobbies & interests: Bouldering, board games, hiking, photography, music, cooking, philosophy, literature, game development.